

Question Bank — Algorithms + Software Engineering

Module 6 companion | PYQ-style + Practice | exam-tagged, answers & explanations

■ How to use this bank

Two sections: PYQ-style (modelled on real IBPS/SSC/GATE/PSU/DSSSB questions) and **Practice** (full coverage by sub-topic).

Every question has answer + explanation. **Tags:** IBPS • Coal India • DSSSB • All 3 • GATE-level • SSC/other.

For IBPS: complexities, sorting/searching, SDLC & models are enough; GATE-tagged (recurrences, algorithm tracing) are Coal-India depth.

SECTION A — PYQ-style (modelled on actual exam questions)

Complexity & searching

Q1

All 3 exams

Big-O notation expresses the:

- (a) Best case
- (b) Upper bound (worst case)
- (c) Average only
- (d) Exact time

Answer: (b)

Big-O gives an asymptotic upper bound; Ω gives lower bound; Θ gives tight bound.

Q2

All 3 exams

SSC/other

Which growth rate is the most efficient (slowest-growing)?

- (a) $O(n)$
- (b) $O(\log n)$
- (c) $O(n^2)$
- (d) $O(2^n)$

Answer: (b)

Order: $O(1) < O(\log n) < O(n) < O(n \log n) < O(n^2) < O(2^n)$.

Q3

IBPS SO IT

All 3 exams

Binary search has time complexity:

- (a) $O(n)$
- (b) $O(\log n)$
- (c) $O(n \log n)$
- (d) $O(1)$

Answer: (b)

Binary search halves the search space each step $\rightarrow O(\log n)$; it needs a sorted array.

Q4

All 3 exams

Linear search worst-case complexity is:

- (a) $O(1)$
- (b) $O(\log n)$
- (c) $O(n)$
- (d) $O(n^2)$

Answer: (c)

Linear search may scan all n elements $\rightarrow O(n)$ worst case.

Q5

All 3 exams

Binary search requires the array to be:

- (a) Unsorted
- (b) Sorted
- (c) Circular
- (d) Linked

Answer: (b)

Binary search only works on a sorted array.

Sorting

Q6

All 3 exams

Which sorting algorithm has $O(n \log n)$ in ALL cases?

- (a) Quick sort
- (b) Bubble sort
- (c) Merge sort
- (d) Insertion sort

Answer: (c)

Merge sort is $O(n \log n)$ best/average/worst. Quick sort is $O(n \log n)$ avg but $O(n^2)$ worst.

Q7

IBPS SO IT

All 3 exams

Quick sort's worst-case time complexity is:

- (a) $O(n \log n)$
- (b) $O(n)$
- (c) $O(n^2)$
- (d) $O(\log n)$

Answer: (c)

Quick sort degrades to $O(n^2)$ with poor pivots (e.g. already-sorted input).

Q8

All 3 exams

SSC/other

Which sorting algorithms use the divide-and-conquer technique?

- (a) Bubble & Selection
- (b) Merge & Quick
- (c) Insertion & Bubble
- (d) Selection & Insertion

Answer: (b)

Merge and Quick sort both use divide-and-conquer.

Q9

All 3 exams

Bubble, Selection and Insertion sort all have average complexity:

- (a) $O(n)$
- (b) $O(n \log n)$
- (c) $O(n^2)$
- (d) $O(\log n)$

Answer: (c)

These simple comparison sorts are $O(n^2)$ on average.

Q10

Coal India MT

GATE-level

Which sort is NOT stable?

- (a) Bubble sort
- (b) Insertion sort
- (c) Merge sort
- (d) Quick sort

Answer: (d)

Quick sort (and selection/heap) are not stable; bubble, insertion and merge are stable.

Design techniques & graphs

Q11

All 3 exams

Dijkstra's, Prim's and Kruskal's algorithms use which technique?

- (a) Dynamic programming
- (b) Greedy
- (c) Backtracking
- (d) Divide-and-conquer

Answer: (b)

All three are greedy algorithms.

Q12

IBPS SO IT

All 3 exams

The knapsack (0/1) and longest-common-subsequence problems typically use:

- (a) Greedy
- (b) Dynamic programming
- (c) Backtracking
- (d) Brute force

Answer: (b)

These have overlapping subproblems → solved efficiently by dynamic programming.

Q13

All 3 exams

Which graph algorithm finds a Minimum Spanning Tree?

- (a) Dijkstra
- (b) Kruskal
- (c) BFS
- (d) DFS

Answer: (b)

Kruskal (and Prim) find the MST. Dijkstra finds shortest paths.

Q14

All 3 exams

SSC/other

Which algorithm finds shortest paths from a single source (non-negative weights)?

- (a) Kruskal
- (b) Prim
- (c) Dijkstra
- (d) Bellman-Ford

Answer: (c)

Dijkstra: single-source shortest path with non-negative weights. Bellman-Ford handles negatives.

Q15

Coal India MT

GATE-level

Floyd-Warshall computes:

- (a) Single-source shortest path
- (b) All-pairs shortest paths
- (c) MST
- (d) Max flow

Answer: (b)

Floyd-Warshall (dynamic programming) computes all-pairs shortest paths.

Q16

Coal India MT

GATE-level

The recurrence $T(n)=2T(n/2)+O(n)$ (merge sort) solves to:

- (a) $O(n)$
- (b) $O(n \log n)$
- (c) $O(n^2)$
- (d) $O(\log n)$

Answer: (b)

By the Master theorem, $T(n)=2T(n/2)+O(n) = O(n \log n)$. GATE-depth.

Q17

All 3 exams

N-Queens is a classic example of:

- (a) Greedy
- (b) Backtracking
- (c) DP
- (d) Hashing

Answer: (b)

N-Queens tries placements and backtracks on conflict.

Software Engineering — SDLC & models

Q18

All 3 exams

The correct SDLC order is:

- (a) Design-Requirements-Coding-Testing
- (b) Requirements-Design-Coding-Testing-Deployment-Maintenance
- (c) Coding-Design-Testing
- (d) Testing-Coding-Design

Answer: (b)

SDLC: Requirements → Design → Coding → Testing → Deployment → Maintenance.

Q19

IBPS SO IT

All 3 exams

Which SDLC phase is usually the longest and costliest?

- (a) Requirements
- (b) Design
- (c) Coding
- (d) Maintenance

Answer: (d)

Maintenance (post-release fixes/enhancements) is typically the longest, costliest phase.

Q20

All 3 exams

SSC/other

Which model is linear with no going back to a previous phase?

- (a) Agile
- (b) Spiral
- (c) Waterfall
- (d) Prototype

Answer: (c)

Waterfall is strictly sequential — best when requirements are fixed and clear.

Q21

DSSSB TGT

All 3 exams

Which model best suits rapidly changing requirements?

- (a) Waterfall
- (b) Agile
- (c) V-model
- (d) Big-bang

Answer: (b)

Agile handles change well via short iterative sprints and customer collaboration.

Q22

Coal India MT

GATE-level

Which model emphasises risk analysis in each iteration?

- (a) Waterfall
- (b) Prototype
- (c) Spiral
- (d) V-model

Answer: (c)

The Spiral model performs risk analysis each loop — suited to large, high-risk projects.

Q23

All 3 exams

The document produced by the requirements phase is the:

- (a) DFD
- (b) SRS
- (c) UML
- (d) ERD

Answer: (b)

The Software Requirements Specification (SRS) states what the system must do.

Design & testing

Q24

All 3 exams

Good software design aims for:

- (a) High coupling, low cohesion
- (b) Low cohesion, high coupling
- (c) High cohesion, low coupling
- (d) High both

Answer: (c)

Goal: high cohesion (focused modules) and low coupling (few interdependencies).

Q25

IBPS SO IT

All 3 exams

Testing a single module in isolation is:

- (a) Integration testing
- (b) Unit testing
- (c) System testing
- (d) Acceptance testing

Answer: (b)

Unit testing checks one module/function in isolation.

Q26

All 3 exams

SSC/other

Black-box testing focuses on:

- (a) Internal code paths
- (b) Functionality without seeing code
- (c) Compiler output
- (d) Hardware

Answer: (b)

Black-box tests functionality via inputs/outputs; white-box examines internal logic.

Q27

All 3 exams

Verification asks 'are we building it right?'; validation asks:

- (a) Is it fast?
- (b) Are we building the right thing?
- (c) Is it cheap?
- (d) Is it secure?

Answer: (b)

Verification = meets spec; validation = meets user needs.

SECTION B — Practice (full syllabus coverage for revision)

B1. Complexity

Q28

All 3 exams

Θ (Theta) notation represents:

- (a) Upper bound
- (b) Lower bound
- (c) Tight bound
- (d) No bound

Answer: (c)

Θ gives a tight bound (upper and lower match).

Q29

All 3 exams

Which is the worst (fastest-growing) complexity?

- (a) $O(n^2)$
- (b) $O(2^n)$
- (c) $O(n \log n)$
- (d) $O(n!)$

Answer: (d)

Order at the top end: $O(n^2) < O(2^n) < O(n!)$.

Q30

All 3 exams

Accessing an array element by index is:

- (a) $O(1)$
- (b) $O(\log n)$
- (c) $O(n)$
- (d) $O(n^2)$

Answer: (a)

Direct indexing is constant time $O(1)$.

Q31

GATE-level

Coal India MT

The Master theorem is used to solve:

- (a) Sorting stability
- (b) Divide-and-conquer recurrences
- (c) Graph colouring
- (d) Hashing

Answer: (b)

The Master theorem gives asymptotic solutions to divide-and-conquer recurrences.

B2. Sorting & searching

Q32

All 3 exams

Which sort repeatedly selects the minimum and places it at the front?

- (a) Bubble
- (b) Selection
- (c) Insertion
- (d) Merge

Answer: (b)

Selection sort selects the min each pass — $O(n^2)$, not stable.

Q33

All 3 exams

Heap sort's time complexity is:

- (a) $O(n)$
- (b) $O(n \log n)$
- (c) $O(n^2)$
- (d) $O(\log n)$

Answer: (b)Heap sort is $O(n \log n)$ in all cases (not stable).**Q34**

All 3 exams

Which sort is best for nearly-sorted small data?

- (a) Quick
- (b) Insertion
- (c) Selection
- (d) Heap

Answer: (b)Insertion sort runs near $O(n)$ on nearly-sorted data and is simple for small inputs.**Q35**

All 3 exams

Interpolation search improves on binary search when data is:

- (a) Unsorted
- (b) Uniformly distributed & sorted
- (c) Linked
- (d) Hashed

Answer: (b)

Interpolation search estimates position; efficient on uniformly distributed sorted data.

B3. Techniques & graphs

Q36

All 3 exams

Greedy algorithms make:

- (a) Globally optimal choices always
- (b) The locally-best choice at each step
- (c) Random choices
- (d) No choices

Answer: (b)

Greedy takes the locally-best option each step; optimal only for certain problems.

Q37

All 3 exams

Which technique stores solutions to overlapping subproblems?

- (a) Greedy
- (b) Divide-and-conquer
- (c) Dynamic programming
- (d) Backtracking

Answer: (c)

Dynamic programming memoises overlapping subproblem results.

Q38

All 3 exams

BFS on an unweighted graph finds:

- (a) MST
- (b) Shortest path (fewest edges)
- (c) All-pairs paths
- (d) Topological order

Answer: (b)

BFS finds the shortest path in terms of edge count on unweighted graphs.

Q39

GATE-level

Coal India MT

Which handles graphs with negative edge weights?

- (a) Dijkstra
- (b) Bellman-Ford
- (c) Prim
- (d) BFS

Answer: (b)

Bellman-Ford handles negative weights (and detects negative cycles); Dijkstra cannot.

Q40

All 3 exams

DFS can be used to detect:

- (a) Shortest path
- (b) Cycles in a graph
- (c) MST weight
- (d) Hash collisions

Answer: (b)

DFS detects cycles and produces topological ordering (on DAGs).

B4. Software engineering

Q41

All 3 exams

Requirements are classified as functional and:

- (a) Optional
- (b) Non-functional
- (c) Temporary
- (d) Hidden

Answer: (b)

Functional = what it does; non-functional = performance, security, usability, etc.

Q42

All 3 exams

Which model pairs a testing phase with each development phase?

- (a) Waterfall
- (b) V-model
- (c) Agile
- (d) Spiral

Answer: (b)

The V-model maps each dev phase to a corresponding test phase.

Q43

All 3 exams

Integration testing checks:

- (a) One module
- (b) Modules working together
- (c) User acceptance
- (d) Hardware

Answer: (b)

Integration testing verifies that combined modules interact correctly.

Q44

All 3 exams

Which testing is performed by/for the client to accept the system?

- (a) Unit
- (b) Integration
- (c) System
- (d) Acceptance

Answer: (d)

Acceptance testing confirms the system meets user requirements before sign-off.

Q45

GATE-level

Coal India MT

Cyclomatic complexity measures:

- (a) Lines of code
- (b) Number of independent paths in code
- (c) Team size
- (d) Cost

Answer: (b)

Cyclomatic complexity counts independent execution paths (test-effort indicator). GATE/PSU-depth.

Q46

Coal India MT

GATE-level

COCOMO is a model for:

- (a) Testing
- (b) Effort/cost estimation
- (c) Database design
- (d) Network routing

Answer: (b)

COCOMO (Constructive Cost Model) estimates software effort/cost.

Coverage & next

✓ What this bank covered

PYQ + Practice across Algorithms (complexity, searching, sorting, design techniques, graph algorithms) and Software Engineering (SDLC, models, design, testing).

GATE-tagged items (Master theorem, negative-weight paths, cyclomatic complexity, COCOMO) are Coal-India / harder-PSU depth.

Note: original questions modelled on real exam patterns, not verbatim reproductions.

This completes the six shared-core module banks (CN, OS, DBMS, Prog/DS, Digital Logic+COA, Algorithms+SE).